

Batch Acceptance Test BAT-09 - Rev B

1. Objective

Verify a complete automatic batch cycle using the effective setpoints and approved return routing. The test is primarily a controller/architecture acceptance test; the WaterNaCl property package may be replaced by StandardWater for topology debugging, but final composition-based acceptance uses the composition-capable configuration.

2. Initial conditions

Item	Initial condition
Simulation time	0 s
B1 composition	w_NaCl = 0.000
B2 composition	w_NaCl = 0.250
B3-B7	Near empty; level ≤ 0.01 m except unavoidable residual volume
Ambient pressure	101325 Pa
Ambient temperature	293.15 K
All automated valves	Closed
P1, P2	Off
B5 heater / B6 cooler / B7 cooler	Off
K1 cooling-water flow proof	Available, ≥ 0.10 kg/s when heater demanded

3. Acceptance sequence

Check	Acceptance criterion
B3 water prefill	Step1 exits at LIS-301 ≥ 0.13 m
B3 recipe	Step2 exits at QI-302 ≥ 0.080 kg/kg
B3 empty	Step3 exits below 0.01 m
B5 charge	Step5 exits at LIS-501 ≥ 0.18 m
Evaporation	Step6 exits at QIS-502 ≥ 0.180 kg/kg
B6 cooling	TIS-602 ≤ 20 degC before P2 return
B7 cooling	TIS-702 ≤ 25 degC before P1 return
Routing	B6 condensate returns to B1; B7 concentrate returns to B2
Parallel join	Cycle reset cannot occur until both branches complete
Cycle restart	Initial becomes active only after process time > 2500 s

4. Logging

Log at 5 s or faster: state/region, all tank levels, QI-302, QIS-502, TI-503, TIS-602, TIS-702, FIS-801, P1/P2, heater/cooler commands and all 15 automated valve commands.